

4 Devices of Articulation: Ubiquitous Computing and Agonistic Collectives

At first glance, the *Spore 1.1* project appears to be an ungainly and puzzling assortment of stuff collected together—a small rubber tree that is surrounded by computer circuitry, wires, and tubing, set atop a reservoir of water, and encased within a transparent plastic cube (Easterly and Kenyon 2004) (figure 4.1).¹ What is this all for? The simplest answer is an automated system for tending a plant. But it is also something more. The design of the system includes a wry twist that shifts it from just an automated system for tending a plant to a system or assemblage that provocatively models a series of associations and dependencies and provides another example of how design can do the work of agonism. The tree is a particular plant, not in species but in origin: it was purchased from the retailer the Home Depot. The significance of this is that the tree comes with the guarantee that if it dies in the first year, the Home Depot will replace it free of charge. The design of the system links the survival of the plant to the Home Depot. Each week, the computer collects stock price information for the Home Depot corporation via the Internet. Whether the plant is watered or not is determined by the performance of the Home Depot stock. If the stock performs well, the plant is watered. If the stock performs poorly, then no water is given to the plant. If the stock continues to perform poorly, the plant eventually dies from lack of water and is removed from the plastic cube, returned to the local Home Depot, and replaced with another tree provided by the Home Depot corporation free of charge, and the process begins again.

Spore 1.1 is prototypical of ubiquitous computing (ubicomputing). As computational components such as integrated circuits and sensors have become smaller and cheaper, they are increasingly embedded in common objects and scattered throughout the environment. The effect is to imbue the world with computational capacities. So through the design of ubicomputing systems, one may encounter objects such as *Spore 1.1* more

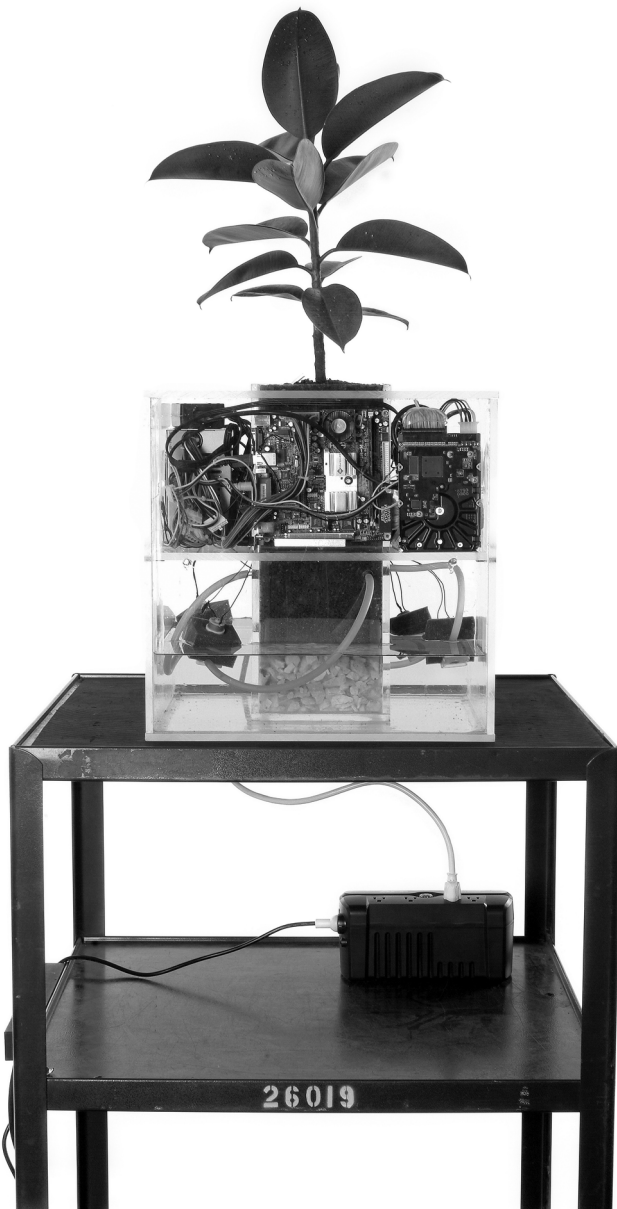


Figure 4.1

Douglas Easterly and Matthew Kenyon, *Spore 1.1* (2007), photo by Luke Hoverman

frequently—computational objects that are defined by their connectivity to other objects and systems and that have the capacity to receive, process, share, and act on data. One consequence of a world that is imbued with computational capacity is that ideas about engaging computation are progressively shifting. Computation is no longer limited to familiar notions of computers. This, in turn, affects the practices and products of design, opening them to a wide assortment of materialities to manipulate with computation, vastly expanding the possible bits and pieces of computation.

This shift should also provide opportunities for distinctive forms and subjects of adversarial design. As the design of ubicomp systems strings together objects and people in various arrangements of exchange and interaction, in what ways will these designs do the work of agonism? What political issues will be evoked by these novel conglomerations of objects, people, and computation? *Spore 1.1* provides hints at answers to these questions. The project establishes simple linkages between objects of varying scales, to surprising effect. These linkages are made significant in a dramatic way as the life or death of a plant is established as being dependent on—connected to—the relative health of a corporation. Although the design of system is provocative, a political perspective is hard to discern. It could be interpreted in multiple ways, for instance, as being about the cyclic and cynical nature of capitalism and consumption, or the responsibilities of corporations to the environment, or our categorizations of nature and technology. But one would be on dangerous ground to assign any single particular political stance or issue to the project. Rather, *Spore 1.1* should be interpreted as suggesting the agonistic potentials of ubicomp by the way it works to produce an articulation of a corporation, its products, consumers, digital networks, financial networks, and the ties that bind them. By establishing these linkages, the design of *Spore 1.1* creates a collective of sorts that people can participate in to consider and question the components of this system and their relations. It is not so much that *Spore 1.1* addresses any one political theme but that it instantiates a model that expresses all of those themes (and others that one might want to project into it) on a scale that is accessible and able to be experienced. Thus, in a manner that typifies agonism, the design of *Spore 1.1* evokes political issues without resolving them. The design of the system identifies the factors at play and establishes their relationships and possible consequences, but it leaves open the space of interpretation and contest. Moreover, it does so in a manner that leverages distinctive qualities of

computation, suggesting that ubicomp does offer opportunities for unique forms of political expression.

Ubiquitous Computing as a Category of Computational Objects

Ubiquitous computing constitutes yet another category of computational objects with distinctive design challenges and possibilities. In its most basic form, ubicomp is about embedding computation into everyday objects, thereby enabling those objects to sense, process, and respond to the actions of others and the surroundings. When these objects are networked together, they are able to share data among each other, resulting in systems and environments of aware and responsive objects—and making computation ever-present. This has the effect of transforming the experiences of these everyday objects and of computers or computation. It also has the effect of transforming how one does design with computation and provides new topics and tactics for political engagement and expression. Specifically, ubicomp brings to the fore the capability to link together computational objects, and that capability can be used to enact politically provocative associations among people, objects, spaces, and actions.

The origins of ubicomp can be traced to research at Xerox PARC in the late 1980s. There, computer scientists, engineers, designers, and social scientists worked together to explore the possibilities and possible effects of embedding computation into objects and dispersing these objects throughout environments. For instance, one of these early technologies was referred to as “tabs.” These were essentially small networked computers that could be worn or carried and that would enable all sorts of interactivity among people, objects, and the environment: “doors open only to the right badge wearer, rooms greet people by name, telephone calls can be automatically forwarded to wherever the recipient may be, receptionists actually know where people are, computer terminals retrieve the preferences of whoever is sitting at them, and appointment diaries write themselves” (Weiser 1991, 80). So from the beginning, ubicomp was positioned as a paradigm of interaction and design that was fundamentally different from familiar notions of using computers. In fact, the discourses of ubicomp are usually framed as a departure from computers as they are commonly known. Ubicomp pioneer Mark Weiser (1991, 94) expressed this sentiment in the opening lines of his canonical article “The Computer for the 21st Century” when he stated, “The most profound technologies are those that disappear. They weave themselves into the fabric of everyday life until they are indistinguishable from it.” With ubicomp, this does not mean that

computation goes away but that the predominance of the computer as a discrete object diminishes as computation becomes part of every object.² As a result, modes of interaction and experiences with computation shift from those mediated by keyboard, mouse, and screen to new forms mediated by objects such as tables, chairs, picture frames, coffee cups, teapots, and jewelry.³ This framing of ubicomp as a fundamentally different experience of computation continues today. Phrases such as “ambient intelligence” and the “Internet of Things” work to separate the idea and activities of computing from those beige, silver, or black boxes one thinks of when one thinks of computers and instead prompt visions of computation as distributed, pervasive, and integrated with the stuff of the everyday.

Since the inception of ubicomp the late 1980s, examples of these technologies and their associated capabilities have expanded beyond the walls of research centers as they have become integrated into consumer products. Although ubicomp may not be a common phrase in everyday discussions, experiences with products that suggest the potentials of ubicomp are becoming more commonplace. Consider the Ambient Umbrella,⁴ one of a line of products from Ambient Devices, a company that works to bring pervasive information technologies into the home. The Ambient Umbrella is composed of an embedded microchip that receives weather forecasts via a wireless Internet connection and a light-emitting diode (LED) embedded in the handle of the umbrella. Through a wireless connection, the umbrella receives a data feed from accuweather.com, which transmits regularly updated weather information culled from a global network of weather stations and satellites. If the forecast is for rain or snow, the umbrella handle pulsates blue light to warn the owner and remind him to take the umbrella along when he ventures outside. This integration of computational sensing and expressive capacities with an everyday object is relatively simple. But when the umbrella is imbued with these computational capacities, it is transformed from an accessory for keeping dry to a computational information display.

The example of the Ambient Umbrella is useful because it shows how ubicomp combines multiple computational qualities and design practices. As a computational designed object, the Ambient Umbrella is a kind of embodied information design that uses the network as a medium of storage and exchange to gather data and then procedurally renders that data in a physically instantiated form. In combining multiple computational qualities and design practices, new kinds of objects and practices emerge. One might not consider an umbrella a computer, but the Ambient Umbrella is a computationally enabled thing. Thinking of umbrellas or other everyday

objects as computationally enabled things is relatively novel terrain in design. As the technical issues of embedding computation into everyday objects are resolved and the industry practice shifts from engineering challenges to design possibilities, design practitioners and design scholars need to ask, What does it mean to do design with computation in the context of ubicomp?

Connectedness and Collectives

Ubicomp draws from and weaves together multiple qualities of the medium of computation, including embodiment, procedurality, transcoding, and the network as a medium of storage and exchange. But the quality that distinguishes ubicomp is connectedness. By *connectedness*, I mean the capacity of ubicomp to link together objects and systems. This connectedness is of a particular kind. It is not limited to a one-to-one connectivity or to connectedness between same or similar objects. Rather, the quality of connectedness that distinguishes ubicomp is a capacity for one-to-many or many-to-many connectivity among an array of diverse objects.

Consider again *Spore 1.1* and the Ambient Umbrella. Both of these comprise a compilation of dissimilar objects—plants, water, pumps, Ethernet cables, umbrellas, LEDs, and microprocessors. What distinguishes ubicomp is not the form or material of any single object. For example, the tree used in the *Spore 1.1* is materially like any other rubber tree (and could be replaced with any other plant from the Home Depot), and the Ambient Umbrella is like any other umbrella in material and form. What distinguishes these products is their connectedness as objects to other objects and systems—an umbrella to a data feed, which is gathered from a network of meteorological monitoring stations, radars, and satellites; and a plant to an automated watering system, the automated watering system to a data feed, parsed from a more general feed of stock market data, which could be seen as a response to a global financial network of events, actions, and desires.

In both of these examples, the ubicomp system is composed of a multiplicity of assorted objects, which are to varying degrees imbued with computation to exchange and express data. In designing the products of ubicomp, the fundamental activity is one of discovering and establishing compelling modes of computational exchange and expression between objects: the design of ubicomp is the design of connectedness. More than just exchange and expression between objects, this connectedness extends

outward to enroll people, other entities in the environment, and even the environment itself.

Most of the discourses of ubicomp focus on the disappearance of the computer, but I focus here on the emergence of collectives. Computation that is distributed throughout and shared among objects draws people and things together into novel arrangements. These novel arrangements of people and things can be understood as collectives—collections and orderings of exchanges, dependencies, resources, and responses among objects, people, and the environment—focused around a particular issue or activity. More than just establishing connectedness, the design of ubicomp results in the formation of these collectives: the design of ubicomp produces collectives. For some, it may seem odd to speak of collectives comprising both people and object. But this activation of objects and their environments and this mingling of a multiplicity of diverse things together provides ubicomp with distinctive political potential. As designer and theorist Julian Bleeker (2009, 173) puts it, “Whereas the Internet of Non-Things was limited to human agents, in the Internet of Things objects are also active participants in the creation, maintenance and knitting together of social formations through the dissemination of meaningful insights that, until now, were not easily circulated in human form.”

Devices of Articulation

If ubiquitous computing is characterized by connectedness and results in the construction of collectives, how does this figure into a discussion of adversarial design? In what ways can these connected collectives function as political provocations, and what political issues are at stake in these collectives? The answers to these questions are found in the notion of articulation. It is my contention that the products of ubicomp should be understood as *devices of articulation*. In the context of adversarial design, these devices of articulation do specific political work: they engage in the formation and expression of *agonistic collectives*. They do so by leveraging the capacities of ubicomp to establish linkages among objects, people, and actions to create open, interpretive, and participatory spaces of contest, thereby providing yet another example of design doing the work of agonism.

This concept of devices of articulation combines the idea of articulation from social and political theory with the more vernacular usage of the term. The common reference point for articulation in social and political theory is the work of Antonio Gramsci (1971), who tied articulation to

class struggle. Just as they did with the notion of hegemony, Ernesto Laclau and Chantal Mouffe (2001) reconsider articulation as a process that extends class struggle and thereby broaden the use of articulation as a theoretical construct. As a general concept in contemporary theory, articulation describes the linkage of discourses and practices to produce hybrid expressions of ideologies and identities. As Laclau and Mouffe (2001, 105) state, articulation is “any practice establishing a relation among elements such that their identity is modified as a result of the articulatory practice.” Articulation thus includes a broad range of activities and contexts that extends the common frames of politics and the political. For example, cultural theorist Dick Hebdige (1981) uses the notion of articulation to describe the ways that subcultures form by mixing styles, attitudes, and values between each other and across class boundaries. Whether the example is one of subcultures or political agendas, articulation is a process that creates “chains of significance” (Laclau and Mouffe 2001; Smith 1998) or connections between discourses and practices and that establishes new meaning, value, and consequences among what otherwise would be disparate, perhaps even incongruent elements.

In his research on articulation work in organizations, sociologist Anselm Strauss (1988, 1993) spoke of the mechanics of articulation. This was an apt reference to the most basic conception of articulation—as a quality of physical systems. In physical systems, *to articulate* means to form a joint, and something that is articulated has multiple parts united by flexible joints that allow for movement. A familiar example of articulation can be found in the human arm and hand. In the arm, three joints—shoulder, elbow, and wrist—articulate a series of bones. Each of these joints, together with an assemblage of muscles and tendons, allows for a range of motion. The hand is articulated through fifteen joints and twenty-seven bones that enable grasping and release actions. Taken together, the articulation of the bones in the arms and hand enables a range of diverse activities, including throwing balls, chopping onions, rolling dice, and writing. Remove any one linkage, however, and the possibilities of motion and action change dramatically. Articulation is also used in engineering to describe the design and use of joints to enable movement in any system of parts, usually with the purpose of enabling a particular functionality that would otherwise be impossible. A familiar example of such articulation is the turntable ladder fire truck. To reach fires at great heights, fire trucks need long ladders, and to carry long ladders, the trucks themselves need to be long. But the longer the truck, the less maneuverable it is. To solve that problem, the turntable ladder fire truck was designed to be articulated with a pivot point that

allows the rear end of the truck to be maneuvered separately from the front of the truck, decreasing the truck's turning radius and thereby better enabling the truck to move safely through streets and neighborhoods. The turntable ladder fire truck thus illustrates the central notion of articulation in physical systems as the making of connections between parts, and these connections are significant because they bind together the parts to establish particular capacities and modes of action.

In making the claim for understanding the products of ubicomp as devices of articulation, I am drawing equally from these two notions of articulation. Articulation in physical systems (and specifically in engineering) is not just a metaphor for political articulation. My intent is to merge these concepts and to claim that articulation in the engineering sense can be, by design, an actual instantiation and form of political articulation. As devices of articulation, the products of ubicomp join together, by design, multiple elements in a manner that transforms the identity and meaning of those elements and results in a new object—an articulated collective.

The idea of an articulated collective is partially drawn from the work of Bruno Latour. In *The Politics of Nature*, Latour (2004) develops the idea of the collective as a way of reconceiving relations among human and non-humans. For Latour (2004, 238), the term refers not to a singular thing, not to a collective, but rather to a "procedure for *collecting* associations of humans and nonhumans." This move to bring and bind together humans and nonhumans is part of a larger project of Latour's to reconsider the roles of tools and machines, animals, laws, infrastructure, and the environment—that is, to reconsider the role of things other than people in the construction of facts and society. For Latour, this is important in order to better describe how things get made and done in the world—not by human hands alone but by networks of actors and actants (nonhuman actors) that result in different configurations and experiences of agencies and effects. Articulation, for Latour, is a quality of such collectives. As he states, "We shall say of a collective that it is *more or less articulated* in every sense of the word: that it speaks more, that it is subtler and more astute, that it includes more articles, discrete units, or concerned parties, that it mixes them together with greater degrees of freedom, that it deploys longer lists of actions" (Latour 2004, 86).

Through my concept of devices of articulation, I want both to focus and extend Latour's ideas. First, my attention is on collectives that are the products of design, and more specifically, designed collectives in the context of ubicomp. Second, in addition to claiming that collectives are articulated, I argue that collectives actively engage in the process of

articulation. As devices of articulation, the products of ubicomp both enable and participate in the ongoing endeavor of establishing linkages between elements in the collective. This idea is not foreign to Latour, for whom objects are assertive and actor networks form to enable or thwart all manner of interactions and effects. But the scale that I want to emphasize and explore is that of the collective as a distinct designed thing, which is more than a singular object but less than a network.

With regard to its political potential, articulation is not by definition agonistic.⁵ Articulation is agonistic when the product of the articulatory transformation questions, challenges, or offers alternatives to dominant perspectives and practices. Put another way, what is agonistic is not necessarily the process of articulation but the outcome of that articulation—the kind of collective created and the affordances of that collective for experiences of contest. Within the frame of adversarial design, *the tactic of articulation constructs linkages between objects, people, and actions that transform them into an agonistic collective—an open space of contest in which the elements gathered together are able to act out a plurality of conflicting practices, values, and beliefs*. These agonistic collectives extend beyond people and discourses alone to gather together all manner of objects, including plants, animals, software, hardware, networks, tables, chairs, buildings, streets, and cities.

The question of precisely what and how the products of ubicomp articulate is explored in the rest of this chapter. To begin with, although ubicomp systems are in part physical systems, they are not articulated in a manner identical to an arm or a fire truck. Articulation in ubicomp systems is not just the introduction of a physical pivot joint between two objects. Rather, one form of articulation in the products of ubicomp comes through sensors, actuators, software, protocols, and networks. That is, articulation is an outcome of leveraging the qualities of procedurality and embodiment, together with the materialities of the object itself, to provide new functionalities, meanings, and possibilities for action by establishing novel linkages between elements in a system. This provides new opportunities and affordances for doing the work of agonism.

Articulating Actions and Ethics

As issues concerning climate change have become more pressing and more public, new technologies and policies have been proposed and in some cases implemented. One such technology-policy hybrid is carbon sinks and carbon offsets. Carbon sinks are natural or artificial reservoirs of chemical compounds: they sequester carbon from the environment and are

proposed as one form of a carbon offset, with the general idea that production and utilization of carbon sinks might counterbalance excess carbon in the atmosphere, thereby mitigating its effects on the climate. Carbon offsets generally operate on an economic model whereby nations or corporations are allocated a certain number of carbon allowances. When the nation or corporation exceeds its allowance, it must purchase or trade for more. The funds from the purchase of additional allowances are then allocated toward projects that allay the excess carbon production elsewhere—for example, investment in renewable energy, energy efficiency, or the production of more carbon sinks. This notion of mitigating energy use through systems of monitoring and exchange extends from the scale of nations and corporations down to communities and individuals. Increasing numbers of tools and resources exist to calculate and monitor individual and local ecological impact, or one's "carbon footprint."⁶ Usually, however, these individual and group systems are designed to focus on the documentation of energy usage or ameliorative actions, and they bypass the political issues and conditions of climate change and control technologies, policies, and practices.

Natural Fuse by Usman Haque (2009) is a ubicomp system that enables users to explore and engage in, on a microscale, the political issues that swirl around carbon sinks, carbon offsets, and the associations of actions and ethics in energy consumption.⁷ Through the design of a ubicomp system, it articulates energy consumption, individual desires, and community consequences to produce an agonistic collective that enlists users as participants in a conflictual model of consumption. This model of consumption mirrors the structure of game theory problem of the prisoner's dilemma in the context of energy usage: a limited amount of energy is made available to a distributed group of participants, who must make individual decisions regarding how much energy they will use—decisions that come with consequences to others in the system.

The *Natural Fuse* project consists of a suite of household plants or plant units—assemblages of a plant, sensors, actuators, and software connected together via the Internet (figure 4.2). These plant units are designed to function as carbon sinks and energy regulators. They are plugged into a home electric outlet and operate as a gateway to the electricity from that outlet. Users can plug an appliance such as a lamp into the plant unit to access electricity from the outlet. The appliance can then be turned on and used but only for as much time as matches the amount of carbon offset by the plant. That is, the capacity to use the appliance is modulated by the capacity of the plant to equalize the carbon cost of the using the



Figure 4.2

Haque Design + Research, *Natural Fuse* (2009)

appliance. This amount of time is extremely short, from a few seconds to a few minutes, depending on the electrical requirements of the appliance.

In the proposed scenario of use, the plant units are distributed among users, who may be close to one another (say, in a single city) or distributed across the globe. A given user comes home and turns on a lamp connected to the plant unit. After a minute or two, the lamp is automatically shut off, having expended its allotment of electricity, and the user faces a choice. She encounters the dilemma.

The plants are connected to each other via the Internet, so each plant unit is a node in a network. By leveraging that connectivity, any user and plant unit can take power allocations from another user or plant unit. Each plant unit is outfitted with a switch marked Off/Selfless/Selfish (figure 4.3). If a user wants to use more electricity than is available from her plant unit, she sets the plant unit to the Selfish setting, and it takes energy allotments from other plants connected to the network. If a plant unit takes too much power, however—if it draws more power than is available in reserve across the network of plant units—then it kills another plant.



Figure 4.3

Detail, Haque Design + Research, *Natural Fuse* (2009)

As a designed object, *Natural Fuse* exemplifies the general concept of ubiquitous computing: it is a system of multiple everyday objects embedded with computational capacities and networked together to exchange data and interact with one another. If we trace the design of the system, by calling out the elements and their connections, we can begin to understand the connectedness that characterizes ubicomp and how the design of that connectedness can foster the production of agonistic collectives.

Each plant in *Natural Fuse* is instrumented with sensors that produce and monitor data and a microprocessor that manages the plant locally. A photoresistor functions as a light sensor, and a humidity sensor is placed in the soil of the plant. These two sensors provide general information about the status of the plant to the microprocessor. The humidity sensor is also connected to a pump, which is connected to a reservoir of water. A valve on the reservoir opens and closes in response to data provided by the humidity sensor to hydrate the plant. There is also a sensor to detect the power draw from any appliance that is plugged in to the plant unit. The data are registered by the local microprocessor and communicated via the Internet to a central database that logs data for all plants in the

Natural Fuse network. The local microprocessor runs an application that compares the electrical draw from the appliance to the amount of the plant's carbon offset.⁸ If the electrical draw exceeds this limit, the local application sends a request to an application on the server requesting more power allocations. The application on the server scans the network of plant units to find available plants to draw power from and sends permission to allocate more power back to the initiating unit, which then opens the circuit to allow the given appliance to power on. If this allocation of power results in the death of another plant—if there is not enough standing reserve of power across the network—then the central application sends an email to all parties involved informing them of the death of the plant. A tally is kept of the number of deaths. Once a plant has reached its three-death limit, a command is sent from the application on the server to the local microprocessor to open a valve on a jar of vinegar attached to the plant unit, dispersing the vinegar into the soil.

As shown by this description, the connectedness of the system extends the computational components of the *Natural Fuse* (figure 4.4). It comprises an articulated collective of people, plants, appliances, electricity, sensors, actuators, microprocessors, software, and vinegar. As a device of articulation, *Natural Fuse* links these elements together in a manner that transforms the individual identity and meaning of each object as it is folded into the collective and transformed. With this transformation, each object acquires new political significance as affordances, dependencies, and responsibilities are established by the design and use of the system. For the appliance to function, it depends on electricity, and the availability of electricity depends on the capacities of a plant unit or a series of plant units. The capacity of the plants to serve as carbon sinks is largely determined by plant biology. The user is responsible for determining under what conditions to allocate electricity to the appliance and when and to what extent to draw electricity beyond the capacities of her plant unit. All of the users in the network link together as a system of both resource and control with finite limits. There is only limited capacity for carbon sinking within the total system, and there is only limited agency available to any one user to make or defend against demands for allocations of electricity.

The collectives of *Natural Fuse* also materialize a series of problematic relations between desires, actions, and consequences and thereby functions as an open, interpretive, and participatory space of contest—and it is in this sense that the collective can be considered agonistic. By leveraging computational capacities for connectedness, the design of *Natural Fuse*

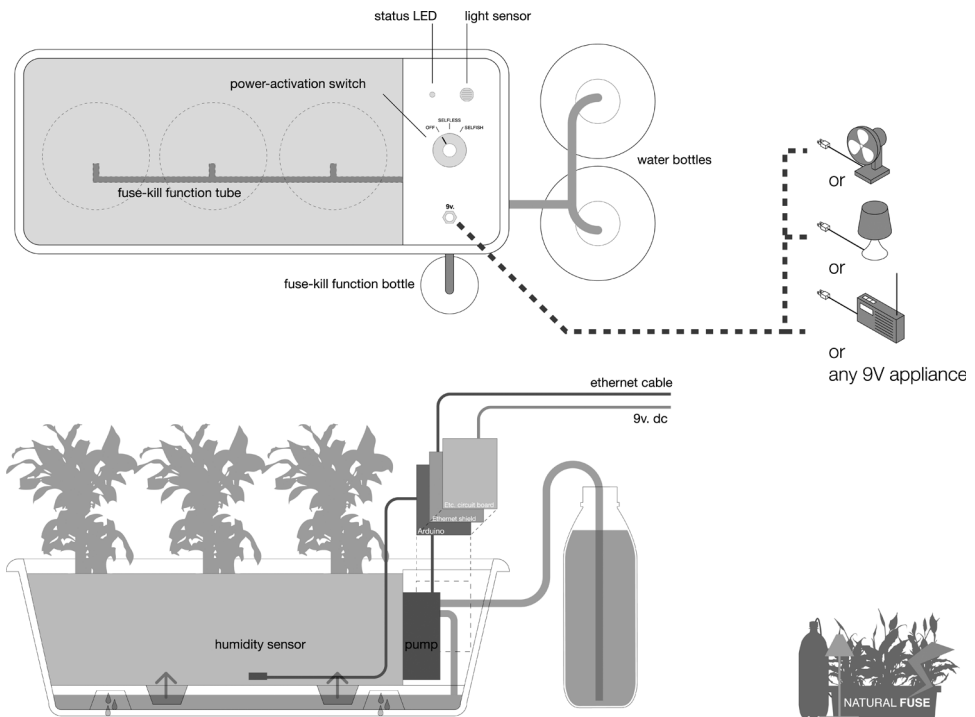


Figure 4.4

System diagram, Haque Design + Research, *Natural Fuse* (2009)

models and allows users to participate in exploring relationships between individual needs and desires and the notion of a common welfare with regard to mitigating climate change. In using the system, the same situation is presented to each participant: Does she take the electricity that she wants when she wants it, or does she find some point of compromise with others, ostensibly for the greater good of all? Does she choose to contribute energy to the collective whole by setting the switch on her plant unit to Selfless, or does she demand energy from others by setting her switch to Selfish? This choice between self-interest and the communal good is the crux of the prisoner's dilemma. But unlike the prisoner's dilemma, the situation is not abstract. It is grounded in a contemporary issue and a specific set of practices—climate change, carbon sinks, and carbon offsets.

Most forms of carbon offsets strive to achieve a state of equilibrium within the system and lead users toward more equalizing behaviors, but *Natural Fuse* enacts an adversarial stance through the design of a system

that is predisposed to disequilibrium. A tendency toward strife is built into the operation of the system. By design, the system enables and almost requires users to engage in contests with one another. In fact, it is not clear whether the counterbalance sought by most carbon sink and carbon offset programs is even achievable with *Natural Fuse*. Each plant unit provides a minimal carbon-sink capacity, and any use of any appliance requires the user to switch to the Selfish mode. In using and reflecting on the use of *Natural Fuse*, one is also prompted, by design, to consider issues of environmental ethics through a micromodeling of the dynamics of resource allocation and the relations between personal desires and actions and the notion of the good of a larger community.

The model and experience provided by *Natural Fuse* is an aestheticized diagrammatic rendering of environmental ethics. The look and feel of *Natural Fuse* make it seem more like an interactive IKEA display than a scientific experiment exploring cooperative behaviors or an engineering prototype for a system of energy monitoring and exchange. But it is precisely this aestheticized quality that lures users into the experience of use, enticing users to participate together with the system in an exploration of this particular issue.

Some may argue that design is more than aesthetics. But the aesthetics of design, in a formal and traditional sense, still have significance in evoking the political. *Natural Fuse* provides an example of the way that pleasing shapes and seductive materials can be leveraged to produce an experience through which an issue is made accessible and of interest by means of a compelling form. With *Natural Fuse*, the user is not asked to pledge alignment to one side of an issue or make sense of cap and trade regulations. Instead, users are invited to take home a curious, seemingly well-crafted, attractive object and use it. The open perspective on environmental ethics that is enacted through *Natural Fuse* provides an outline of the issue that users can fill in with their own actions and interactions and thereby participate in the articulation of the issue through the use of the system.

So, *Natural Fuse* does not take a normative or prescriptive stance on environmental ethics. It does not punish those who consume to the detriment of others, nor does it reward them, nor does it reward those who operate their appliances within the limits of their capacity for offset. Indeed, the openness of the system makes it possible to read a number of potential ethical perspectives into the project. As a device of articulation, *Natural Fuse* gives material and interactive form to a series of relations, highlights the ethical issues at stake, and prompts users to engage—to

participate—in a model of political interaction. It does not resolve the situation of resource allocation and usage. Instead of using design as a means of providing a solution, it uses design to problematize the situation.

Articulating Countercollectives

The *Natural Fuse* project provides an example of ubiquitous computing in the home, but ubicomp is not limited to domestic environment. Urban environments—cities—are also lively sites of ubicomp research and design. In part, this is due to increasing numbers of wireless data networks within cities. In fact, many of the advances and problems of ubicomp have emerged in tandem with the increased availability and capacity of wireless data networks. These networks provide access to various devices that enable the exchange of data between users, devices, and the applications running on them. Although these urban wireless networks provide access, they are not all open spaces. They are regulated and monitored channels of transmission and communication. Whether the access is provided by a municipality or a commercial vendor, the data that flow across the network are subject to the laws and ordinances of the providers, and open to their surveillance. The extent of covert electronic surveillance of network communications is unknown. The notion of electronic surveillance, however, is now part of the public consciousness. We hear about governments that monitor online terrorist chatter, mail that is intercepted in national or corporate espionage cases, and media companies that track, identify, and prosecute those who download or distribute copyrighted materials.

The *Ad-hoc Dark (roast) Network Travel Mug* by Mark Shepard (2009a) is intended to address issues of network surveillance by providing an alternative means of network connectivity and exchange (figure 4.5).⁹ This ubicomp system allows users to surreptitiously send short text messages to one another across a closed network, using travel mugs that are input devices, displays, and network routers in disguise. Through its design, the system articulates a collective of resources that enable users to avoid, even momentarily, network surveillance systems without having to forgo the uses of networks for simple social communication.

The functionality of the system is based on a design that, like *Natural Fuse*, illustrates the fundamental notion of ubicomp: at its core, it is a design of connectedness achieved by embedding computational technologies into everyday objects and then networking those objects together. The primary component of the *Ad-hoc Dark (roast) Network Travel Mug* is a common aluminum mug, the kind used to keep drinks warm as one travels



Figure 4.5

Mark Shepard, *Ad-hoc Dark (roast) Network Travel Mug* (2009a)

to and from work, school, or play. What is different about this mug is that it functions as node within an ad-hoc mesh network—a particular kind of digital network in which each node can function as a router, transmitting data to other nodes along the network. As the coffee mugs/nodes move through space, the network changes in size and capacity as users carrying the mugs come into and drift out of range of one another. When enough *Ad-hoc Dark (roast) Network Travel Mugs* are near each other, they form a closed network of data transmission and exchange. This network is “dark” not because it is related to coffee but because it is run as a clandestine parallel network that is outside of the reach of electronic monitoring by municipal or commercial networks.

Each mug is outfitted with a wireless ad-hoc mesh network module, a microprocessor that communicates with other modules via radio frequency (RF), an LCD screen for display, and a microcontroller that runs the necessary software for establishing and maintaining the network and providing basic output via the screen. Software on the network module and microprocessor sends out a repeating signal via RF, which announces its presence and availability to other modules. It also listens for the same signal as it comes from other modules. When one module detects another, a connection is established, and when enough connections are established, a network is formed that allows the transmission and reception of data. When one module leaves the configuration and is no longer within range

of any other module, the network registers that module as unavailable and routes the data among the remaining available modules. In addition, a button embedded on the side of the mug picks up tapping sequences by users. These taps are procedurally transformed into alphanumeric codes, which are transmitted to other mugs within the network, enabling a basic messaging system.

The *Ad-hoc Dark (roast) Network Travel Mug* is part of a larger project by Shepard titled the *Sentient City Survival Kit* (2009c), which uses design to explore the possibilities and consequences of ubiquitous computing in the city. For this project, Shepard has created a series of concept designs and working prototypes. Each concept and prototype offers a novel experience with ubiquitous computing and engages issues of “privacy, autonomy, trust and serendipity in this highly observant, ever-more efficient and over-coded city” (Shepard 2009c). The notion of sentience that Shepard uses to frame his project highlights the ways that computation as a theme is intertwined with contemporary discussions of the city. This is not surprising. Cities have always been shaped by and reflexively shaped technology, from the Roman aqueducts and arcades of Paris to the highways of Los Angeles and Atlanta. Today the city as a “networked society” and as “spaces of flows” is influenced by the qualities of computational technology.¹⁰ The space, place, and experience of the city are now shaped by information and communications technologies infrastructure and services, from cell towers to Internet cafés. New topologies of a city can even be based on degrees of data access and data service usage. Likewise, the qualities of the city reflect back on technology research and design, evidenced by the invention of phrases such as “urban computing” and “urban informatics,” which apply perceived qualities of the urban to computational media, products, and services.¹¹ In such design domains, distinctions are made in the design of computational systems for use in the city versus the rural, suburban, or exurban.

Just as with the data networks themselves, the physical space of the contemporary networked city is a space of pervasive monitoring. In fact, nowhere has the idea of the networked city been more fully realized than in systems of urban surveillance. Networked communications surveillance of the kind that the *Ad-hoc Dark (roast) Network Travel Mug* seeks to counter is one such mode of surveillance. But perhaps the most common mode is video surveillance using charge-coupled device (CCD) cameras—digital systems that produce images without the use of film or tape, usually recording the image data to hard drives. In their simplest forms, contemporary networked surveillance systems combine hundreds or even thousands of

CCD cameras distributed throughout a city to provide a glimpse of the city as a whole and a record of the city to be searched and referenced. So pervasive is this surveillance that in some cities, such as London, it is unusual for people to be out of the view of a CCD camera. More advanced tracking and monitoring systems use a variety of sensors to produce finer-grained data and a heightened awareness and documentation of the city. For example, the ShotSpotter Gunshot Location System recognizes and locates gunshot sounds by using a network of audio sensors distributed throughout the city.¹² Through these and other distributed sensing systems, the city becomes a feed of digital data as sounds, air quality, traffic flow, and other factors are observed, registered, and recorded. The data can be used to initiate real-time responses or be stored and analyzed to serve as the basis for some action in the future.

As these monitoring and tracking systems continue to be developed, they will continue to affect how we perceive and live in the networked city. Geographer Stephen Graham has theorized this confluence of technology and urbanism since the 1990s, and his work is influential in the cultural studies of ubicomp and design. In “Sentient Cities: Ambient Intelligence and the Politics of Urban Space,” Michael Crang and Stephen Graham (2007) theorize that society may be moving toward a notion of “anticipatory” cities. In an anticipatory city, sensors and video cameras are distributed throughout the urban environment and are combined with data-mining and pattern-recognition techniques so that users can predict the behaviors of individuals and groups of people.

Shepard himself references Crang and Graham in his *Sentient City Survival Kit* project (Shepard 2009c), and his *CCD-Me Not Umbrella* (2009b) can be taken as an attempt to address and counter pervasive video surveillance and the use of computational vision systems as part of that surveillance.¹³ The *CCD-Me Not Umbrella* is a standard umbrella with an array of infrared LEDs studding the umbrella canopy, powered by a stack of batteries housed in the handle (figure 4.6). These lights are not there to illuminate the environment but rather to distort the scene as it is perceived by CCD cameras and to confound the computational vision algorithms that are used for object detection in surveillance systems.

Imagine that users want to avoid CCD surveillance, either because they have something to hide or perhaps because they oppose the practice of surveillance. As they walk down the street carrying the umbrella, they turn the umbrella lights on and off repeatedly. This generates visual effects that are perceptible only to CCD systems (distortion in the captured scene) and frustrate the process of object detection, thereby thwarting effective



Figure 4.6

Mark Shepard, *CCD-Me Not Umbrella* (2009b)

surveillance. A single person equipped with a *CCD-Me Not Umbrella* could thus counter individual monitoring, while a crowd of users with *CCD-Me Not Umbrellas* might compound the effects, resulting in images of city streets that are nearly unrecognizable (figures 4.7 and 4.8).

Although the *CCD-Me Not Umbrella* is not a computational artifact, it is designed to interface with a computational system—that is, as a component of a ubicomp system, albeit a disruptive component. Its design is based on an astute understanding of the technical capacities and limitations of computation and the ways that other objects and their qualities (such as LEDs and the infrared wavelength of light) might combine to work together with or, in this case, against computational technologies. As a common feature of computational vision systems, object detection and tracking uses algorithms to read an array of pixels in a digital image and from that reading identify and follow particular shapes. Through these systems, it is possible to isolate individuals from a crowd, distinguish among people, automobiles, and buildings, and even track and analyze facial expressions, gesticulations, and gait. The array of infrared LEDs in the *CCD-Me Not Umbrella*, however, would frustrate many algorithms used



Figures 4.7 and 4.8

Mark Shepard, *CCD-Me Not Umbrella* (2009b). These images show a computational vision system attempting to monitor the umbrella. The umbrella is progressively disrupting the image so that the person is no longer singled out. The software is confused and is tracking light traces in the image.

within these systems. Repeatedly turning on and off the infrared lights could affect the registration of pixels that are stored in system memory and that separate the foreground (the individual who is surveilled) from the background. As a result, the computational vision system would lose track of the individual because it would be unable to separate him or her from the total scene.

As a device of articulation, the *CCD-Me Not Umbrella* produces what I call a *countercollective*—a collective that operates in a manner contrary to another collective. For example, the capacities, resources, and qualities brought together in the design of the *CCD-Me Not Umbrella* work to undo the capacities, resources, and qualities of many advanced video surveillance systems. In this way, countercollectives exemplify and engage in a particular task of agonism that Mouffe (2007) refers to as “disarticulating the existing order.” In both a discursive and a material sense, countercollectives work by unhinging the joints that bind another collective together. In doing so, the countercollective disables or thwarts the capacities of that other collective. The collective produced by the *CCD-Me Not Umbrella* is articulated for the purpose of disarticulating the collective of the surveillance system. This notion of articulating a countercollective provides one more example of the distinctive political potential of ubicomp that is made possible by leveraging qualities of connectedness and the interrelated dependencies that characterize connectedness.

Just as a series of dependencies can be traced through *Natural Fuse*, so too can the unhinging or disarticulation be traced through *CCD-Me Not Umbrella*. This unhinging can be understood as a series of dependencies undone by the design of *CCD-Me Not Umbrella*: the light emitted by LEDs on the umbrellas makes the video collected from the cameras unreadable by the computational vision systems, and thus the computational vision systems are unable to provide the monitoring and tracking they were intended for. The *CCD-Me Not Umbrella* thus provides an example of an agonistic encounter between systems or collectives in which one collective does the work of agonism by attempting to disarticulate another collective. Returning to the dual notion of articulation as both an engineering and political endeavor, the *CCD-Me Not Umbrella* exemplifies articulation as something more than just a discursive maneuver. The disarticulation that the *CCD-Me Not Umbrella* attempts to perform occurs on a material level by leveraging the material design qualities of one collective against the material design dependencies of another. Here articulation in the engineering sense is also an instantiation and form of political articulation.

The Pluralism of Collectives

Ubicomp systems employ more than computation and the affordances of individual objects such as coffee mugs and umbrellas. There is a pluralistic quality to these collectives. These systems extend outward to draw in and combine a much greater breadth of people, technologies, and objects than might be initially thought. Through articulation, the design of ubicomp systems link and incorporate the material and social environment into collectives. Through this, they become essential elements of the design. To fully realize the political potentials of ubicomp requires recognizing the extent of this social and material enrollment and the ways that ubicomp systems can provide novel spaces for agonistic expression and engagement.

Consider again the *Ad-hoc Dark (roast) Network Travel Mug* and the *CCD-Me Not Umbrella*. Both of these systems depend on elements and phenomena that extend beyond the obvious parts of umbrellas, coffee mugs, LEDs, and microprocessors. The design of the *Ad-hoc Dark (roast) Network Travel Mug* includes more than the instrumented travel mug. The design also includes the subway car, the habits and customs of morning and evening commuters, and the spatial arrangement of people in that space and at that time. This collection of factors is foundational to the design of the system and requisite for the system to operate. With the *Ad-hoc Dark (roast) Network Travel Mug*, to establish a network among nodes, the nodes must be present and near each other—that is, the system requires particular spatial arrangements. Subway cars provide a bounded space in which that arrangement can emerge. But more than just the material construction of the subway car, the design also leverages the conditions of use of the subway—the close distribution of people within subway cars during the morning and evening commutes. Taken together, these qualities of the subway car and commute provide the material and social infrastructure of use and experience. A similar pattern can be seen in the design of the *CCD-Me Not Umbrella*: its design includes an umbrella studded with LEDs, city streets, and the software that it works to disrupt. Even the weather figures into its design, since weather is the excuse for using the umbrella.

In both cases, elements and phenomena of the social and material environment are made into integral aspects of the systems. They are articulated by design into the systems. Recognizing that ubicomp collectives articulate a host of elements into their design, beyond the computational objects themselves, provides important insights into how ubicomp can do the work of agonism. Like any manner of articulation, the combination of

constituent elements transforms their identity in the production of a new collective with new possibilities for action and meaning. As previously noted, what makes a collective agonistic is the extent to which it produces an open space of contest where conflicting values and practices can be acted out. These spaces of contest are the contexts of use that are transformed by agonistic articulation. That is, when designed or interpreted from an agonistic perspective, ubicomp products can transform the context of use into places and events in which political contest might unfold.

For example, with the *Ad-hoc Dark (roast) Network Travel Mug* and the *CCD-Me Not Umbrella*, riding in a subway car during the morning commute and walking on a rainy city street are no longer just moments spent traversing the city. They become places and events for engaging, challenging, and providing alternatives to urban life as it intersects with ubicomp. The subway car and the commute become a context in which to work around systems of surveillance and engage in parallel, “dark” modes of communication. Likewise, the rainy city street becomes a context in which to work against systems of surveillance and engage in minor acts of disruption. And with *Natural Fuse*, the domestic environment and everyday activities as banal as turning on a lamp become transformed into places and acts in which questions of energy consumption are experientially confronted and the simple act of using an appliance is politicized, made into a situation with political meaning and consequence. Although the previous qualities and associations between these elements and phenomena remain—the subway during the morning commute is still a crowded place, rain is still wet, appliances still require electricity—through the articulation by design, new political associations and possibilities are constructed.

One effect of broadly articulating the material and social environment is that the situations and actors of the political are greatly expanded. By including elements and phenomena of the social and material environment into the design of a ubicomp system and as part of an agonistic collective, those elements and phenomena become politicized in ways they were not before. Who would have previously considered a subway car a space for engaging issues of network surveillance or a houseplant a thing for engaging issues of energy consumption, and individual actions and desires? Computation and the visions and practices of ubicomp seem to amplify the potential for political engagement with objects by enabling them to be linked to and link others into associations that have political meaning and significance.

In this sense, ubicomp systems actively participate in articulation, drawing together and giving new meaning to the constituent elements of

a system, regardless of what those elements are. The ubicomp system engages those elements and phenomena, connects them through a series of design dependencies and exchanges of data and, in the process, transforms them toward new ends. And there seem to be few limits to what can be articulated into a ubicomp system. Nearly any site or object of any scale can become a site or object politicized by design by leveraging the quality of connectedness provided by ubicomp. Just as ubicomp embeds computation in everyday objects distributed throughout the environment, so too can it, by design, draw out the political in everyday situations and practices.

At its core, articulation is a transformative activity, and with ubicomp systems, that transformation is enacted, in part, beyond the designer and the user. Ubicomp systems actively gather, order, and express beyond the actions and intentions of the designers and users. The transformative event of articulation is not solely assigned to human actors. In this way, we are returned to or at least turned toward Latour's project of broadening the understanding of how action happens beyond human effort, taking into account the capacities of objects.

This design of connectedness that characterizes the design of ubicomp also returns us to and mirrors earlier discussions of procedural authorship. With procedural authorship, designers construct rules for representations, those rules are executed in software, and depending on the variables (including user input) and the data available, different representations result. The adversarial design of ubicomp is similar, but rather than producing representations, it produces spaces for contest. With ubicomp systems, the designer constructs a series of linkages leveraging multiple qualities of computation, including procedurality and embodiment, and when those linkages are enacted, depending on the actions of users and the qualities of the elements articulated into the design, different experiences result. In regard to political expression, this suggests that a set of conditions can be gathered together, but the message itself cannot be easily scripted. Indeed, this is the opportunity of adversarial design in the context of ubicomp—to set stages and provide props for the open exploration, perhaps even discovery, of the political significance and meaning of social issues and conditions.

Summary

Ubiquitous computing provides yet another example of a category of computational objects with distinctive design challenges and opportunities for

doing the work of agonism. Doing design with ubicomp means establishing and structuring the connectedness of the constituent elements of a ubicomp system. The range of these elements is arguably boundless because in the grand vision of ubicomp set out by researchers, any object can be imbued with computational capacities and distributed throughout any environment. In the near future, aware and responsive networked objects and spaces will create environments of seemingly sentient things, which will interact with people and with each other in lively fashion. Regarding the political, ubicomp systems have the potential to do the work of agonism through processes of articulation by using the quality of connectedness to construct agonistic collectives. These collectives are political and agonistic because they enable users and objects to participate together in making, exploring, and contesting alternatives to a wide variety of societal issues and conditions.

Although the design of contemporary commercial ubicomp systems, such as the Ambient Umbrella, begins to fulfill the early promises of ubicomp by imbuing everyday objects with computation, most such examples are still relatively staid designs. The collectives they construct continue and extend familiar and common relations. For example, with the design of the Ambient Umbrella, all that has occurred is that two points in a series of relations and dependencies have been combined: news of inclement weather is now available through the object used to mitigate inclement weather. Users do not have to listen to weather reports to learn this information and then reach for the umbrella; they can reach for the umbrella when it communicates news of the weather. This is clever, perhaps even useful, but it is not really a compelling demonstration of the possibilities of ubicomp for prompting new relations between people, objects, and issues and for instigating new forms of action. Moreover, the design of the Ambient Umbrella does not address the political issues or potentials of ubicomp. One function of systems such as *Spore 1.1*, *Natural Fuse*, and the products of the *Sentient City Survival Kit* is to demonstrate alternatives to the mainstream practices and discourses in ubicomp design and product development. As demonstrative alternatives, these products operate in a manner similar to the robots discussed in chapter 3. They function as exemplars of things, situations, opportunities, and consequences that might be. Like Kelly Dobson's *Blendie* and *Omo* (2007a), which challenge us to reconsider the assumptions and purposes of social robotics, projects such as *Spore 1.1*, *Natural Fuse*, and the *Sentient City Survival Kit* challenge us to reconsider the contexts and implications of ubicomp. Across all of these cases, these designed objects do more than just level a critique. They

provide a material demonstration of possible alternatives and shift debate from a discursive format to an experiential form. With *Natural Fuse*, users are not abstractly arguing about the ethics of energy consumption, cooperation, and conservation but are participating in a model of it. And with *Blendie*, users are not discussing robots and emotion at a distance but rather directly engaging a machine in an affectively charged encounter.

In fact, articulation can be considered to be a tactic that is akin to reconfiguration but with distinctions of scale. As a tactic, articulation tends to work across a range of objects that are greater in difference and distribution than is the case with reconfiguration. In some cases, this span of distribution and difference is merely a matter of definition of what constitutes an object or a collective. For example, a challenging boundary case would be a collection of robots working together, perhaps as a swarm working toward a singular goal, or perhaps a group of robots that work together with scientists and a host of other devices, such as the Mars rover robots. But more often, this difference of scale is immediately apparent. For example, *Blendie* consists of an instrumented kitchen mixer and single user, but *Spore 1.1* spans from the individual consumer to the multinational corporation, from the biological health of a single plant to the economic health of the Home Depot.

Perhaps most significant for both the design scholar and the practicing designer is to understand that through the process of articulation—by forging connections between objects, people, space, and actions—the identities and meanings of each element are transformed. A plant is not just a plant, a coffee mug is not just a coffee mug, an umbrella is not just an umbrella. Through design, they are now objects that people can use to engage in questioning and contestation to provoke and probe political issues and relations. This transformation extends beyond the objects to include the social and material context of use and user. A user is now a politicized actor. In these agonistic ubicomp systems, the user engages in questioning and contestation as she decides what amount of electricity to use at what time, whether to participate in the construction of a parallel data network or in the disruption of surveillance systems, or whether and how to intervene in the care of a plant. Through the design of agonistic collectives, one can begin to envisage ways in which users are not only witnesses to adversarial design but also participants in doing the work of agonism.